

A memristor-based adaptive neuromorphic decoder for brain–computer interfaces

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Practical brain–computer interfaces should be able to decipher brain signals and dynamically adapt to brain fluctuations. This, however, requires a decoder capable of flexible updates with energy-efficient decoding capabilities. Here we report a neuromorphic and adaptive decoder for brain–computer interfaces, which is based on a 128k-cell memristor chip. Our approach features a hardware-efficient one-step memristor decoding strategy that allows the interface to achieve software-equivalent decoding performance. Furthermore, we show that the system can be used for the real-time control of a drone in four degrees of freedom. We also develop an interactive update framework that allows the memristor decoder and the changing brain signals to adapt to each other. We illustrate the capabilities of this co-evolution of the brain and memristor decoder over an extended interaction task involving ten participants, which leads to around 20% higher accuracy than an interface without co-evolution.

Brain–computer interfaces (BCIs), which establish a direct link between the brain and an external machine, could help integrate human–machine functions and to diagnose and treat brain diseases. Recently, such interfaces have been used for brain-to-text communication¹, speech synthesis^{2–4}, motor control⁵ and spinal cord stimulation⁶. However, the brain is a complex dynamic system, and brain signals constantly evolve over time and often fluctuate^{7–17}, which could degrade the performance of BCIs. Therefore, a practical BCI should be able to adapt to evolving brain signals and interpret intentions during collaborative interactions.

Current adaptive decoders primarily use data labels generated by preupdated decoders^{1,4,15}, leading to a lack of active brain involvement¹⁸, which impedes the ongoing improvement in BCI performance.

Furthermore, rapid growth in brain–machine links^{19–22}, commands^{23,24} and decoding algorithm complexity^{3,4} creates challenges for conventional silicon chips based on the von Neumann architecture²⁵ to decode the increasing volume of brain signals. In such architectures, to process continuous clips of brain signals in real time, vast amounts of data need to be frequently shuffled between the computing and memory units, leading to increased latency and energy consumption. Therefore, an energy-efficient brain signal decoder and a hardware-friendly update scheme are required to create co-evolutional BCIs.

Neuromorphic computing chips could provide a solution. Neuromorphic hardware has been shown to offer high energy efficiency in various signal processing tasks, including brain decoding for BCIs^{26–29}. In particular, the memristor—a tunable resistive switching device that

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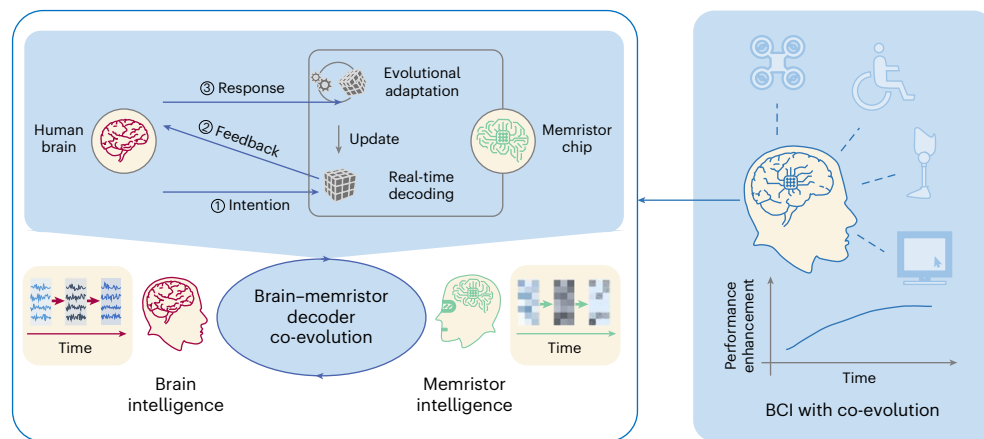


Fig. 1 | Schematic of BCI with brain-memristor decoder co-evolution.

Enhanced BCI that integrates human brain intelligence with memristor-chip-based machine intelligence (memristor intelligence) is enabled by brain-memristor decoder co-evolution. The process involves three key signals: intention, feedback and response. Initially, the brain sends out a control intention through its activities. Subsequently, the memristor chip decodes it, transmitting the decoding result to the brain as feedback. After that, the brain responds

to whether the decoding is correct, guiding the memristor chip update and decoding model transfer. The three sets of evolving multichannel brain signals indicated by red arrows illustrate the fluctuations in brain activities. By contrast, the evolving decoder maps indicated by green arrows illustrate the adaptive memristor decoders. The right panel shows an example of typical BCI-controlled devices, such as a drone, wheelchair, prosthesis and mouse.

mimics synaptic weight—provides a promising platform for neuromorphic computing. The non-von Neumann paradigm of in-memory computing can be implemented in situ on analogue memristor crossbar arrays via fundamental physics laws^{30–32}, and such an implementation can enhance the decoding energy efficiency of BCIs. Moreover, memristor chips support in situ parameter tuning^{33–35}, enabling adaptive decoders with a gradual update capability for continuous brain-machine interaction. There have been attempts to use memristors to decode brain signals, including epilepsy detection^{36–38} and distinguishing motor imagery^{39,40} and neural firing signals^{41,42}. However, these studies mainly relied on prerecorded or simulated brain data using a small number of memristors that do not directly interact with the brain (Supplementary Table 1).

In this Article, we report a neuromorphic and adaptive decoder for BCIs that is based on a memristor chip (Fig. 1). To achieve high accuracy, speed and low energy consumption in decoding brain signals, we develop a one-step memristor decoding strategy. We test the system in a complex real-time task involving brain-controlled drone flight, and show that it achieves decoding performance equivalent to software methods but with 1,643 times lower energy consumption and 216 times higher normalized speed than a conventional central processing unit (CPU)-based system. Furthermore, we develop an interactive update scheme that allows the memristor decoder and the changing brain signals to adapt to each other naturally. We illustrate the capabilities of this brain-memristor decoder co-evolution with ten participants engaged in a co-evolution task, each undergoing around 6 h of experiment. This shows that the BCI with brain-memristor decoder co-evolution achieved around 20% higher accuracy than that without co-evolution.

Design of memristor-chip-based BCI

To construct a prototype memristor-chip-enabled BCI, we used the steady-state visually evoked potential (SSVEP) as the intention, the decoding result from the memristor chip as the feedback and the error-related potential (ErrP) as the response (Fig. 2). SSVEP, closely correlated with visual stimuli^{43,44}, is one of the most used brain signals to build a BCI. ErrP, a neurophysiological signal associated with error processing in the brain^{45–47}, is evoked when the brain recognizes an error. Figure 2a illustrates the framework of the memristor-enabled BCI. The proposed interactive update scheme of the memristor-chip-based decoder is illustrated in the hybrid path. Here ErrPs are used to

accumulate new data in the online interaction process. More specifically, if the memristor decoder's decision does not elicit an ErrP, the corresponding SSVEP signal and the decoding result serve as a new training sample. The decoder undergoes updates during the interaction phase as these new training samples accumulate. We can use the updated decoder for the subsequent loop through this human-in-the-loop mechanism.

Figure 2b shows the memristor-chip-based decoding platform, and the die photo of the memristor chip is shown in Fig. 2a. We elaborate on the chip fabrication process in the Methods. The chip consists of a $1,024 \times 128$ (that is, a 128k-cell) memristor array that supports massively parallel multiply-accumulate computations and peripheral circuits for input and output buffers and drivers. Each cell in the array is in the form of a one-transistor-one-resistor cell. The memristor has a material stack of TiN/TaO_x/HfO₂/TiN, as shown in the cross-sectional transmission electron microscopy image (Fig. 2c). The memristor array exhibits its excellent analogue switching characteristics for high-efficiency in-memory computing (Extended Data Fig. 1).

We used the representative ensemble task-related component analysis (TRCA) algorithm⁴⁸ to decode the brain signals. The non-ideal device characteristics of memristors, such as read noise and limited precision, could introduce errors when performing matrix-vector multiplication computations on the memristor array. Moreover, such errors could accumulate when computing with multiple arrays or steps of matrix-vector multiplications. To efficiently deploy the ensemble TRCA on the memristor chip, here we propose a one-step decoding strategy that considers both accuracy and efficiency by minimizing the error accumulation from sequential computations based on multiple memristor arrays/chips. The key idea is to merge the original three-step operation, including preprocessing, feature extraction and pattern recognition, into a one-step operation (Fig. 2d; Methods provides the mathematical derivation). This method substantially reduces the computation complexity and the required number of memristor arrays. As an example, for a task with 8-channel signals and 12-class control commands, 6.5× fewer computations can be achieved using this one-step strategy compared with the conventional three-step implementation (Fig. 2e). A more detailed analysis on the reduction ratio of computational complexity between the three-step and one-step implementations under various task situations is provided in Supplementary Note 2

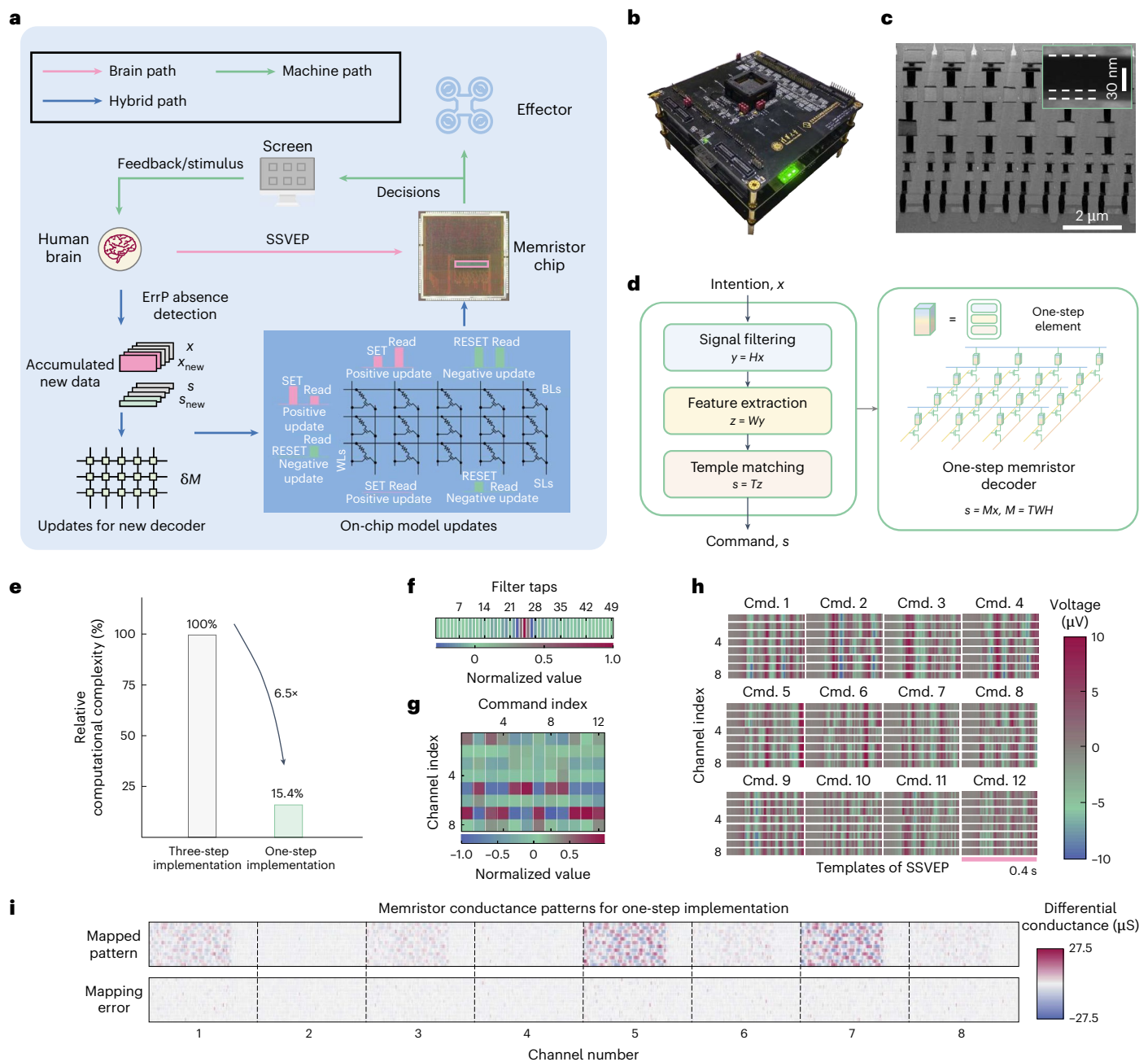


Fig. 2 | Design of a memristor-chip-enabled BCI. **a**, Framework of the memristor-chip-enabled BCI. The pink, green and blue arrows represent the brain, decoder and hybrid paths, respectively. In the brain and decoder paths, the memristor chip decodes the brain's SSVEP signals and then feeds the results back to the brain. In the hybrid path, ErrP responses from the brain are detected to accumulate new online training samples to update the decoder. SET/RESET pulses are then applied on the memristor chip to update the model. SL, source line; BL, bit line; WL, word line. **b**, Photograph of the memristor-chip-based decoding platform. **c**, Cross-sectional transmission electron microscopy image of the memristor chip with one-transistor-one-resistor cells. The inset shows the TiN/TaO_x/HfO₂/TiN material stack of the memristor device. **d**, One-step

implementation of the SSVEP decoding algorithm on the memristor array by merging the original preprocessing (temporal filtering), feature extraction (spatial filtering) and pattern recognition (template matching) into one matrix computation. **e**, Comparison of the computational complexity for the three-step and one-step decoding algorithm implementations for an 8-channel and 12-command drone flight control task. **f–h**, Coefficient maps of the temporal-filtering (**f**), spatial-filtering (**g**) and template-matching (**h**) signals for the conventional three-step implementation in this task. **i**, Conductance maps for the concise one-step implementation. The top panel is the mapped differential conductance on the memristor array, and the lower panel shows the corresponding mapping error.

and Supplementary Fig. 1. The temporal filter, spatial filter and SSVEP template signal for the conventional implementation are illustrated in Fig. 2f–h, respectively. The new strategy combines these components into one matrix and then maps them onto the memristor chip as the device conductance in differential pairs (Fig. 2i). We describe more details about the mapping process in the Methods. Evidently, this

one-step strategy simplifies the operations, thereby avoiding the accumulation of errors caused by multistep operations on the memristor chip. More interestingly, it also increases the interpretability of the decoder, whose pattern could reflect the channel preferences of the brain and the sparsity of control parameters in the interactive BCI tasks, as investigated later. For example, the pattern shown in Fig. 2i

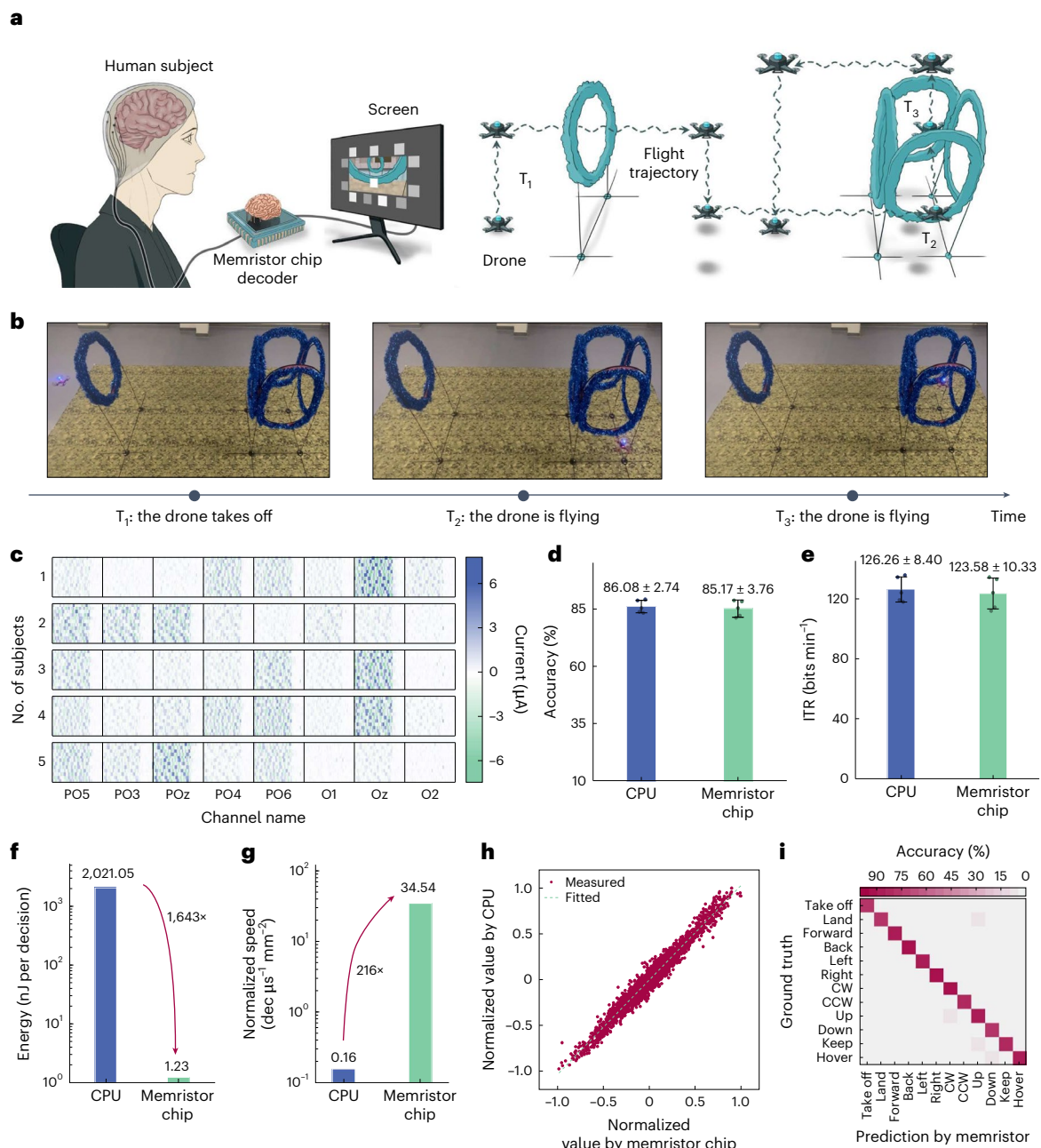


Fig. 3 | Real-time brain-controlled drone flight with a memristor-chip-based decoder. **a**, Schematic of the experimental setup. The left panel illustrates that the subject controls the drone flight through the memristor-chip-based decoder. The screen shows 12 encoded stimuli of flight commands along with a real-time video stream of the drone. The right panel shows the flight scene and trajectory in the online experiment. T_1 – T_3 represent several key time points during the online drone flight. **b**, Key frames of the memristor-chip-based online decoding experiment. **c**, Differential conductance maps of the memristor decoder for

five different subjects. **d–g**, Performance comparisons between the CPU and memristor chip, including accuracy (**d**), ITR (**e**), energy consumption (**f**) and normalized decoding speed (**g**). The dots in **d–e** indicate the corresponding results of different subjects. The error bar shows the mean $\pm$ s.e.m. ($n = 5$). **h**, Comparison of the decoding results from the CPU and memristor chip in the simulated online experiments, showing a linear relationship and high consistency. **i**, Confusion matrix of the memristor-chip-based decoding experiment. CW, clockwise; CCW, counterclockwise.

suggests that the brain relies more on the signals from channels 7, 5, 1 and 3 than the other channels for this specific task, because they have larger parameter magnitudes.

Real-time brain-controlled drone flight

To verify the feasibility of the memristor-chip-based decoder for BCI, we first carried out a sophisticated online task using four-degrees-of-freedom (4-DOF) real-time brain-controlled drone flight (Fig. 3a). Besides the memristor chip, the experimental setup

also includes a brain signal acquisition system (Synamps2, Neuroscan), a drone (DJI RoboMaster TT), a screen showing the flight commands and artificial obstacles to guide the drone flight (Methods, Extended Data Table 1 and Supplementary Figs. 2 and 3 describe the detailed experimental setup). Supplementary Video 1 shows a typical trial in which the drone successfully completed its intended three-dimensional trajectory around obstacles using the memristor-chip-based decoder with a one-step ensemble TRCA. Figure 3b illustrates the key frames of the three flight nodes (T_1 – T_3). A comparison of brain-controlled

schemes with prior works is provided in Supplementary Note 3 and Supplementary Table 2.

To quantify the decoder performance, we performed the memristor-chip-based simulated online experiments on five subjects (marked A1–A5). Figure 3c plots the differential conductance maps of the memristor decoder, indicating that different subjects could have distinct channel preferences. This observation stems from the commonly reported cross-subject variabilities⁸ in BCIs. For instance, subject A1 primarily relies on signals from channels Oz, PO4 and PO6, whereas subject A5 favours channel POz. Figure 3d–e compares the performance of BCIs based on the memristor chip and a conventional decoding hardware, that is, an Intel CPU, which exhibited a similar information transfer rate (ITR; 123.58 ± 10.33 bits min⁻¹ for the memristor chip versus 126.26 ± 8.40 bits min⁻¹ for the CPU) and decoding accuracy ($85.17 \pm 3.76\%$ for the memristor chip versus $86.08 \pm 2.74\%$ for the CPU). Extended Data Table 2 lists the detailed results for each subject. Meanwhile, the memristor-chip-enabled BCI consumed $1,643\times$ less energy and had $216\times$ higher normalized speeds than the CPU-based system (Fig. 3f,g). The benchmark is detailed in the Methods and Extended Data Table 3. Such equivalent accuracies suggest that the decoding results of the memristor chip are in excellent consistency with those of the CPU (Fig. 3h). Figure 3i further presents the confusion matrix of the memristor-chip-based decoding results for different commands. We also compared our memristor-based decoding strategy with a fully connected neural network strategy, which has the same structure with our one-step ensemble TRCA. The memristor-based strategy achieved a much higher accuracy than that of the neural-network-based strategy ($38.83 \pm 10.79\%$; Supplementary Note 4 and Supplementary Table 3). The above results clearly validate the feasibility and advantages of the memristor-chip-based decoder for BCI.

Previous research reported that the sparsity of control parameters clearly exists in many BCI tasks^{7,49}. To investigate the effects of decoding parameters with various magnitudes on the system performance, we carried out parameter-pruning experiments using binary masks (Extended Data Fig. 2). Extended Data Table 4 summarizes the results; the decoders that retained the top 20% parameters in magnitude could achieve similar accuracy ($<1.5\%$ accuracy loss; Extended Data Table 4a) as the unpruned decoders. In comparison, removing only the top 5% of the parameters could result in a substantial degradation in accuracy ($>10.0\%$; Extended Data Table 4b). This result clearly indicated the sparsity of parameters and that decoder parameters with higher magnitudes played dominant roles in decoding the brain signals. We further investigated the importance and preference of different brain recording channels for the specific task. Specifically, we use the model parameters to calculate the average magnitudes for each channel with 100 inputs (a 0.4-s-duration signal with a 250 Hz sampling rate has 100 time points) and 12 outputs. We also calculate the accuracy using each single-channel signals, where the 100 time points in the corresponding channel are used as inputs and the other channels are set to zeros for decoding. These methods are based on the fact that when using a linear model, the magnitudes of the parameters can represent the importance for the specific tasks. Supplementary Fig. 4a shows the average magnitude of parameters in each channel of the decoders. Supplementary Fig. 4b presents the decoding accuracy based on a single-channel brain signal for every subject. Evidently, one or a few individual channels contained most of the critical information for decoding (for example, channel Oz for subject A1, which alone already achieved a high accuracy of 90.0%), whereas the others contributed very little (such as channel PO4 for subject A3, which alone achieved a chance-level accuracy of 8.75%). Comparing the patterns of the average magnitude (Supplementary Fig. 4a) and accuracy (Supplementary Fig. 4b) from a channel-wise perspective, they are found to have a high similarity of 0.783. This is because the channels that achieved higher accuracy often had parameters with larger magnitudes. Thus, they were more important than the other channels in the brain-controlled drone

flight task. The above results showcase the close correlation between the memristor-chip-based decoder and the brain activities from both parameter-wise and channel-wise perspectives, which could offer a useful tool to study the ‘credit assignments’ to brain signals in the task. In addition, the sparse parameters of the decoder are favourable for hardware implementation and system optimization, leading to lower decoding costs in terms of chip area and energy consumption.

Brain–memristor decoder co-evolution for enhanced BCI

To demonstrate the co-evolution of the brain–memristor decoder for an enhanced BCI, we conducted two experiments with the memristor: the first one is a simulated update (SU) experiment, and the other one is a human-in-the-loop update (HLU) experiment. In both experiments, we initiated the decoder by training it with an offline dataset and subsequently updated it with the newly acquired data. Finally, we evaluated the performance of the updated decoder in the extended experiment. The decoding algorithm used in these experiments was also the one-step ensemble TRCA. The major difference between these two experiments was the manner in which new samples were labelled. The SU experiment used ground truths for labelling to validate whether the decoder update improved the BCI’s performance. By contrast, the HLU experiment used ErrPs in online setups for labelling, intending to explore the feasibility of interactive brain–memristor decoder co-evolution.

In the SU experiment (on subject A3), after five updates, the decoding accuracy exhibited a notable improvement of more than 5% with the brain–memristor decoder co-evolution compared with its non-co-evolving counterpart (Extended Data Fig. 3a). This outcome unequivocally demonstrates that the memristor-chip-based decoder, coupled with the interactive update scheme, possesses the ability to adapt to evolving brain signals, thereby enhancing the decoding performance over an extended period. To track the change in SSVEP signals, the decoder map underwent periodic updates, tending to converge to a stable state. This convergence is evident in the correlation coefficients, which progressively increased initially from 0.81 to 1.0 at the conclusion of the experiment (Extended Data Fig. 3b,c).

To further demonstrate the brain–memristor decoder co-evolution, we conducted the HLU experiment on ten subjects (marked B1–B10), in which subject-dependent ErrP detectors were first established to accumulate online data (Methods). To achieve high-accuracy detection, we carried out an ErrP calibration experiment. Extended Data Fig. 4 shows the performance of a typical detector. We studied the effects of the training set size and the algorithm threshold on the ErrP detection performance. Increasing the training set size generally enhanced the performance (Extended Data Fig. 4a). An optimal threshold of about 0.5 achieved the highest balanced accuracy of 82.6% (Extended Data Fig. 4b). Extended Data Fig. 4c,d details the ErrP detection using this threshold. Our next step was to conduct an online ErrP detection using the same workflow, observing how the brain responded to the correct or incorrect decoding result obtained using the memristor chip. When incorrect decoding occurred during the online experiments, an apparent positive spike was observed at around 300–400 ms following the stimulus presentation (Extended Data Fig. 4e).

We further analysed the BCI performance with the memristor-chip-enabled adaptive decoder. Throughout the experiment, the decoding accuracy of the BCI, coupled with brain–memristor decoder co-evolution, consistently outperformed its counterpart without co-evolution across all subjects (Extended Data Fig. 5). Some subjects (such as B5 and B6) even performed with nearly 100% accuracy after co-evolution. These findings confirm that the memristor-chip-based adaptive decoder can maintain and gradually improve its accuracy in real-time scenarios under varying conditions and long-term experiments involving multiple subjects at different dates and times

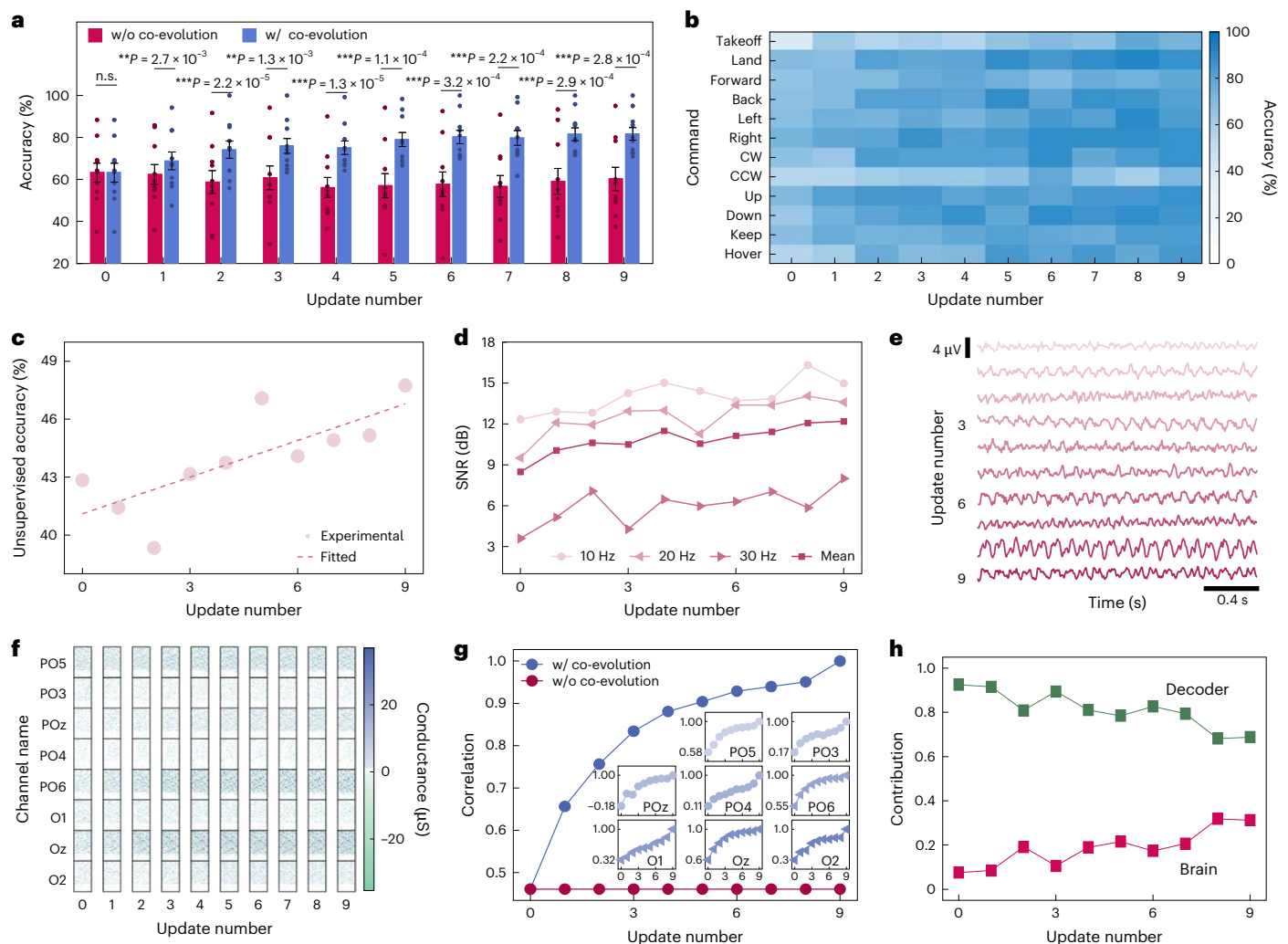


Fig. 4 | Online brain-memristor decoder co-evolution experiments with HLU.

a, Decoding accuracies versus update cycles for BCI with (w/) and without (w/o) co-evolution for ten subjects. The error bar shows the mean $\pm$ s.e.m. ($n = 10$). Two-sided paired-sample t -test: $^{**}P < 0.01$, $^{***}P < 0.001$. **b**, Evolution of the decoding accuracy for 12 different control commands. Each pixel represents the average result of ten subjects. **c**, Accuracies calculated using FBCCA after different updates. Each dot represents the average result of ten subjects, and the dashed line is the result of linear fitting ($R^2 = 0.588$, $P < 0.01$). **d**, Progressive enhancement

in signal-to-noise ratio (SNR) for the task-related components at 10 Hz and its harmonics during evolution. **e**, Temporal waveforms of task-related components after the decoder update. **f**, Adaptive memristor decoder maps after each update cycle for eight recording channels. **g**, Correlation coefficient between the decoder maps after each update and the final one. Insets: correlation coefficient for each channel of the decoder. **h**, Evolution of the individual contribution of the brain ($\alpha/(\alpha + \beta)$; red) and the decoder ($\beta/(\alpha + \beta)$; green) to the enhanced co-evolutionary BCI performance.

(Supplementary Table 4). In particular, the performance gap between the two approaches widened progressively over time. By the ninth update, corresponding to an ~six-hour experiment duration, we observed a remarkable improvement of ~20% in the average accuracy (Fig. 4a). Figure 4b and Supplementary Figs. 5 and 6 illustrate the evolution of the decoding accuracy of each command in BCIs with and without brain-memristor decoder co-evolution, respectively. These figures suggest that the decoder acquired new information during interaction and consistently improved accuracy with each update.

To reveal the continuous fluctuations in brain signals and the change in brain control state for the task, we analysed the evolution of subject-averaged accuracies (Fig. 4c and Methods) using the unsupervised algorithm called filter bank canonical correlation analysis⁴³ (FBCCA). In this analysis, an unsupervised learning algorithm is adopted to isolate the effect of brain signals from that of the decoder in the co-evolution experiments that involve both. The increasing trend shows the gradually improved brain control state during the six-hour-long experiment. Throughout the co-evolution process, the task-related components progressively adopted a waveform that

resembled a combination of sinusoidal signals synchronized with the stimulus frequency. Computational analysis revealed a notable enhancement in the signal-to-noise ratio of these components (Fig. 4d). We also observed the evolution of similarity between the extracted feature components of brain signals and the template signals (Supplementary Figs. 7 and 8). For all subjects, the similarity increased, presenting a trend of stabilization with gradually saturated values. Figure 4e shows the temporal waveforms of brain signal components after every decoder update. These results from all ten subjects illustrate how the task-related brain signals changed during the interaction.

We further analysed the evolution of adaptive memristor decoders. Figure 4f plots the adaptive memristor decoder map of subject B1 in the initial state and after nine consecutive updates, and Supplementary Fig. 9 plots the decoder maps of the other subjects. The evolving pattern for all the subjects showed a trend of stabilization and convergence. To verify this, we calculated the correlation coefficients between the patterns of the present and final decoder maps. For example, Fig. 4g shows the evolution of correlation coefficients for the decoder of subject B1 and each channel (insets), which continuously

increased from -0.74 to 1.00. Supplementary Fig. 10 shows comparable results for other subjects. To study the credit assignments in this task, that is, the importance of different channels and their changes, we then analysed the evolving direction of the decoder by calculating the fitted slope of the parameter magnitudes. Supplementary Fig. 11 plots the slopes of all eight channels for subject B1. Supplementary Fig. 12 shows the evolution of the positions of the top 15% parameters. Different channels can evolve in distinct directions. As an example, channel PO3's decoder map exhibited a downward trend in magnitudes (Supplementary Fig. 13a). By contrast, channel Oz showed an upward trend (Supplementary Fig. 13b). To study the evolution trends for different control commands, here we plotted the distributions of the decoder parameters for each command (Supplementary Fig. 13c for subject B1 and Supplementary Fig. 14 for all subjects). As the update progressed, the decoder parameters became increasingly distributed, as the standard deviations increased from -3.3 μS to -5.8 μS . In addition, we observed no noticeable difference in the update rules for different commands.

Towards practical applications, previous studies suggest that a BCI would achieve optimal performance when both the brain and the decoder can adapt to new scenarios^{75,51}. In line with this, our system demonstrated this dynamic interplay, as evidenced by the brain learning over time and the memristor decoder adapting accordingly. To gain a deeper understanding of the enhanced performance of our co-evolutional BCI, we further explored the individual contributions of these two components. Supplementary Fig. 15a plots the accuracy of the co-evolutional BCI (Fig. 4a, $A_{\text{co-evo}}$), the brain control state (Fig. 4c, A_{brain}) and the memristor decoder's effectiveness (A_{decoder}) for this task (Methods). A linear regression analysis on the z-scores (Supplementary Fig. 15b) revealed that the slope of $A_{\text{co-evo}}$ was greater than that of both A_{brain} and A_{decoder} . In simpler terms, the co-evolutional BCI exhibited faster performance improvements compared with the human brain and the memristor decoder. This accelerated improvement can be attributed to the synergistic evolution of the brain and the decoder, collectively enhancing the performance of the co-evolutional BCI. As a first-order approximation, we can decompose their individual contributions using the following linear relationship:

$$A_{\text{co-evo}} = \alpha A_{\text{brain}} + \beta A_{\text{decoder}},$$

$$\alpha, \beta \geq 0,$$

where α and β denote the fitting coefficients associated with the brain and the decoder, respectively. We can explore the evolution of α and β by fitting this model in each update. Our findings reveal that the contribution of the decoder ($\alpha/(\alpha + \beta)$) decreased whereas that of the brain ($\beta/(\alpha + \beta)$) increased as the evolution progressed (Fig. 4h). This suggests that in the initial stage of co-evolution, the decoder played a dominant role. However, as the co-evolution progressed, the brain gradually learned, leading to a dynamic balance between the decoder and the brain for decision-making. Such a dynamic shift may indicate a complex interaction between the human brain and the decoder, probably influenced by factors^{50,52,53} such as the brain's plasticity, decoder's adaptivity and human-in-the-loop feedback. Unlike conventional decoders, our memristor-chip-based neuromorphic decoder, with inherently adaptive features, holds the potential to revolutionize brain-machine interaction.

Conclusions

We have reported a neuromorphic BCI based on a memristor chip. We developed a one-step memristor decoding strategy, which reduces computational complexity and hardware costs, and serves as a tool for studying credit assignments in subject-dependent BCI tasks by analysing the parameter sparsity of the memristor decoder. The decoder was used to perform a brain-controlled 4-DOF drone flight

task, achieving a software-equivalent accuracy of $85.17\% \pm 3.76\%$ and an ITR of 123.58 ± 10.33 bits min^{-1} . Our approach operates at three orders of magnitude lower energy consumption and two orders of magnitude higher normalized speed than conventional CPU-based systems. We developed brain-memristor decoder co-evolution capabilities with an interactive decoder update scheme based on ErrP, which achieved around 20% accuracy improvement in a six-hour experiment. During this HLU experiment, both brain signals and memristor-chip-based decoder gradually converged to their stabilized states. The decomposition of their individual contributions revealed an increasing role of the brain and a decreasing contribution of the decoder. This dynamic shift indicates the brain's learning and the decoder's adaptivity. Our framework for brain-memristor decoder co-evolution could be generalized to various types of energy-efficient BCI, including invasive recordings such as spikes¹⁴ and electrocorticography⁵⁴, providing long-term stability and adaptive capabilities. It could also potentially be used for medical treatment and rehabilitation, such as closed-loop neural modulation^{55,56}. In addition, by exploring the close interaction and evolution between the biological brain and neuromorphic memristor chip, our approach could potentially lead the development of human-centred hybrid intelligence.

Methods

Memristor chip fabrication and characteristics

Design and fabrication of 128k-cell memristor chip. The chip comprises a $1,024 \times 128$ one-transistor-one-resistor memristor array and Si complementary metal-oxide semiconductor peripheral circuits, including input buffers and drivers to support read/write and computation operations. Each cell in the array consists of an access transistor and a memristor with a material stack of TiN/TaO_x/HfO₂/TiN with dimensions of $0.5 \mu\text{m} \times 0.5 \mu\text{m}$. We fabricated the peripheral circuits and Si transistors in the array using a standard 130 nm Si complementary metal-oxide semiconductor process. The 8 nm HfO₂ layer was deposited by atomic layer deposition to serve as the resistive switching layer, and the TaO_x and TiN layers were deposited by sputtering to serve as the thermal enhanced layer and top/bottom electrodes, respectively.

Memristor-chip-based decoding platform. The platform mainly consists of three customized printed circuit boards: a field-programmable gate array board, a power board and a socket board. The field-programmable gate array board is used for control. The power board generates various voltage signals through the integrated digital-to-analogue converters for the memristor chip mounted on the socket board. Besides the memristor chip, the socket board also contains eight analogue-to-digital converters for array readout.

Mapping the decoder model onto the memristor chip. We used a standard write-verify programming scheme to map the decoder model onto the memristor chip. Each memristor device has two basic steps when mapping it to the target conductance. First, read the device to obtain its current conductance, and calculate the difference between the target and current states. Second, if the difference is positive (negative), a SET (RESET) pulse is applied to the device. The above two steps are repeated until the difference is within a preset margin.

Brain signal recording and decoding

Brain signal recording and preprocessing. In this study, brain signals were acquired using a Synamps2 system (Neuroscan) at a sampling rate of 1 kHz and were resampled to 250 Hz before feeding to the memristor chip. Here 64-channel whole-head signals were recorded. Eight of these channels over the parietal and occipital areas (PO5, PO3, POz, PO4, PO6, O1, Oz and O2) were used as the SSVEP signals. Their 32 channels over the other areas (FP1, FP2, F7, F3, Fz, F4, F8, FC5, FC1, FC2, FC6, T7, C3, Cz, C4, T8, TP7, CP5, CP1, CP2, CP6, TP8, P7, P3, Pz, P4, P8, PO7,

POz, PO8, O1 and O2) were used to represent ErrP-related signals. During the recording, we kept the impedance of all the electrodes below 10 k Ω to ensure good signal quality. The reference electrode was placed at the vertex.

Ensemble TRCA decoding algorithm. Ensemble TRCA is a supervised method and requires user calibration data to train the decoder model⁴⁸. The model contains the individual template χ^{N_f} calculated by averaging the interfrequency trials and the ensemble spatial filter $U_e = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_{N_f}]$, where $\mathbf{u}_k$ represents the spatial filter trained with k th stimulus data. $\mathbf{u}_k$ is obtained by maximizing the covariance among the interfrequency trials. For a testing trial x' , the target is identified by finding the maximum correlation coefficient between $x'U_e$ and $\chi_k U_e$:

$$\mathbf{r}(k) = \rho(x'U_e, \chi_k U_e), \quad (1)$$

where $\mathbf{r}$ is the decision vector of the testing trial x' . $\rho(a, b)$ refers to the Pearson correlation analysis between matrices a and b .

Derivation of one-step decoding strategy on the memristor array

Equation (1) can be viewed as a three-step computation comprising temporal filtering, spatial filtering and template matching. They can be merged into a one-step operation (Supplementary Note 1 shows the details). This process can be briefly viewed as

$$\mathbf{s} = M\mathbf{x}, \quad (2)$$

$$M = TWH, \quad (3)$$

where $\mathbf{s}$ and $\mathbf{x}$ represent the decision and brain signals, respectively. M represents the one-step decoding matrix. T , W and H are the one-step implementation matrices for temporal filtering, spatial filtering and template matching, respectively. To implement it on the memristor array, we can use G , $\mathbf{v}$ and $\mathbf{i}$ to represent M , $\mathbf{x}$ and $\mathbf{s}$, respectively:

$$\mathbf{i} = G\mathbf{v}, \quad (4)$$

where G , $\mathbf{v}$ and $\mathbf{i}$ are the differential conductance matrix, input voltage vector and output current vector, respectively.

Brain-controlled drone flight experiment

Visual stimulus presentation. We presented the visual stimuli using a sampled sinusoidal stimulation method encoded with a joint frequency- and phase-modulation strategy⁴³. The stimulus sequence $\mathbf{s}(f, \varphi, i)$ was generated by modulating the luminance of the screen using the following equation:

$$\mathbf{s}(f, \varphi, i) = \frac{1}{2} \left\{ 1 + \sin \left[2\pi f \left(\frac{i}{RR} \right) + \varphi \right] \right\}, \quad (5)$$

where f and φ are the frequency and phase of the stimulus, respectively; i indicates the frame index in the stimulus sequence; and RR represents the refresh rate of the screen. The value of $\mathbf{s}$ ranges from 0 to 1, where 0 represents dark and 1 represents the highest luminance. In this study, 12 commands for 4-DOF drone flight were encoded to the frequencies and phases of the stimulus (Extended Data Table 1).

Drone actuation. This study develops a first-person-perspective user interface that integrates a video stream from the drone into visual stimuli to enable the control of the drone's flight in four degrees of freedom. The user interface displays 12 stimuli corresponding to 12 flight commands, which include take-off, land, hover, keep, clockwise, counterclockwise and rectilinear motions along six directions. The decoding results of the brain signals are utilized to yield a flight

command. Direct commands, such as take-off, land, hover and keep, are executed immediately. By contrast, the other commands, known as motion control commands (that is, clockwise, counterclockwise and rectilinear motions), are converted into a four-element speed vector $\mathbf{v}_n$. The vector's elements represent pairs of opposite directions, which are forward-back, left-right, up-down and clockwise-counterclockwise, respectively. For instance, if the currently decoded control command is forward, then a unit speed value would be set into the speed vector $\mathbf{v}_n$, that is, $\mathbf{v}_n = [1, 0, 0, 0]$, where n is the speed vector index. The successive m speed vectors will be summed up and determine the movement speed in different dimensions $\mathbf{V}_n$ of the drone:

$$\mathbf{V}_n = \sum_{n=m+1}^n \mathbf{v}_n, (n > m - 1). \quad (6)$$

Offline and simulated online experiments. We conducted an offline experiment to collect training samples for training a user-specific decoder of brain signals. The experiment comprised six blocks, each containing 12 trials corresponding to 12 control commands presented in a random order. Before each trial, we presented a visual cue for 0.5 s that filled the target stimulus with red. Subjects were instructed to shift their gaze to the cue-indicated target as quickly as possible. Following the offset of the visual cue, all stimuli flickered for 2 s, divided into 5 consecutive epochs of 0.4 s each. We conducted a simulated online experiment comprising two blocks to evaluate the decoding performance. All stimuli flickered by repeating the 0.4 s epochs within each block. To acquire the labelled data for evaluating the decoding accuracy, a red-filled circle positioned beneath the simulation square served as a visual cue to guide the subjects. After every ten epochs, the visual cue changed from one stimulus to another, demanding the subjects to gaze at the cue-indicated stimulus. The simulated online experiment comprises 240 epochs, each target being cued twice.

Online brain-controlled drone flight experiment with memristor-chip-based decoder. We set up the online experiment scenario in a spacious room that was well lit, quiet and comfortable. As shown in Supplementary Video 1, a ring-shaped obstacle with a diameter of 0.6 m and a cube with 0.6-m-long edges, supported by their respective bases, were placed on a blanket on the floor. We asked the subject to control the drone to complete a three-dimensional trajectory-tracking task. In the online experiment, the stimuli flicker continuously with 0.4 s epochs, whereas the eight-channel brain signals were decoded in real time through a memristor-chip-based decoder for drone flight control. Throughout the online experiments, we carefully monitored the drone's status to ensure the safety of the subject and devices.

Performance evaluation

ITR and its calculation. The ITR is commonly used to evaluate the performance of the BCI⁴⁴, and we defined it as

$$\text{ITR} = \left(\log_2 [N] + P \log_2 [P] + (1 - P) \log_2 \left[\frac{1 - P}{N - 1} \right] \right) \times \left(\frac{60}{T} \right), \quad (7)$$

where N is the number of commands, P is the accuracy and T is the time taken to make each decision. For the brain-controlled drone flight task, $T = 1.2$ s as three consecutive epochs were used to make one decision. For the co-evolution task, $T = 0.4$ s as one epoch was used to make one decision.

Energy benchmark. A 1×800 vector was used as the input data in the brain-controlled drone flight experiment. The decoding process computed the multiplication of the input vector and the decoder matrix, whose size is 800×12 . This step consumes ~ 1.23 nJ for the memristor

chip (Extended Data Table 3). For a representative Intel Core i9-12900 CPU, whose energy efficiency is about 9.5 GOPS W^{-1} , the above step consumes $800 \times 12 \times 2 \text{ OPS}/9.5 \text{ GOPS } W^{-1} = 2,021.05 \text{ nJ}$. As a result, the energy consumption of the memristor chip was $\sim 1,643\times$ lower than that of the CPU.

Benchmark of normalized decoding speed. Each one-bit input took $\sim 50 \text{ ns}$ on the memristor chip. In the brain-controlled drone flight experiment, with 10-bit input data, the computation time for the task was $50 \text{ ns} \times 10 = 500 \text{ ns}$. For the chip area of 0.0579 mm^2 (Extended Data Table 3), we estimated the normalized decoding speed (that is, the number of decoding decisions per unit time per unit area) as $1/500 \text{ ns}/0.0579 \text{ mm}^2 = 34.54 \text{ decisions } \mu s^{-1} \text{ mm}^{-2}$. For comparison, we still used the Intel Core i9-12900 CPU with a chip area of 200 mm^2 and computing power of 600 GOPS; therefore, the computation time by CPU was $800 \times 12 \times 2/600 \text{ GOPS} = 3.2 \text{ ns}$ for the task. We then estimated the normalized decoding speed as $1/3.2 \text{ ns}/200 \text{ mm}^2 = 0.16 \text{ decisions } \mu s^{-1} \text{ mm}^{-2}$. Consequently, regarding the normalized decoding speed, our memristor chip showed an advantage of $\sim 216\times$ over the CPU.

Brain-memristor decoder co-evolution experiment

ErrP calibration experiment. To train the ErrP detector, we asked subjects to first participate in a calibration experiment to collect the target and non-target brain signals as the training samples. We modified the user interface from the previous drone flight control experiment and used the same basic stimuli settings. The experiment comprised five blocks, each containing 36 trials (that is, going through all 12 stimuli thrice). A 1.5 s break occurred between two consecutive trials, during which we presented a visual cue to guide the subject to switch gaze to the next stimulation. Each trial had ten stimulation epochs, with visual feedback presented after the offset of each stimulation epoch. There was a 20% chance that the visual feedback was inconsistent with the stimulus indicated by the visual cue, constituting an error occurrence. The brain signals collected during error occurrence epochs were labelled as targets. Conversely, the remaining 80% of epochs classified as correct were labelled as non-targets. All the target and non-target trials were collected to train the ErrP detector.

Data preprocessing. The signals collected from the calibration experiment were first re-referenced by averaging the TP7 and TP8 channels and then extracted into trials in the interval $[-400, 600] \text{ ms}$ that was time-locked to the feedback onset. These trials were filtered in $[1, 10] \text{ Hz}$ using a Chebyshev type-I infinite impulse response filter.

ErrP detector training and online ErrP detection. This study used the discriminate canonical pattern matching⁵⁷ to detect ErrP features. The discriminate canonical pattern matching comprises three key components: the construction of discriminative spatial patterns, the construction of canonical correlation analysis patterns and pattern matching. We used the signals as a training set to differentiate between the target and non-target trials. We calculated each class's discriminative spatial pattern spatial filter and template using the training data. Next, we applied these discriminative spatial pattern spatial filters to the test trial and templates to enhance distinguishability after transformation. Then, we used the canonical correlation analysis algorithm to reveal the underlying correlation between the processed test trial and the templates. We determined the test trial's class by comparing its similarities with the training template. We used true-positive rate (TPR), true-negative rate (TNR) and balanced accuracy (BACC) to evaluate the performance of our detector (Extended Data Fig. 4). The first two metrics represented the chances of successful predictions in the target class (error, positive sample) and the non-target class (correct, negative sample), respectively. The BACC metric was used for

evaluating the learning process with class imbalance. These metrics are defined as follows:

$$TPR = \frac{TP}{TP + FN}, \quad (8)$$

$$TNR = \frac{TN}{TN + FP}, \quad (9)$$

$$BACC = \frac{TPR + TNR}{2}, \quad (10)$$

where TP, FP, TN and FN are the number of true-positive, false-positive, true-negative and false-negative samples, respectively.

Long-term brain-memristor decoder co-evolution experiment.

Before conducting the HLU experiment, we conducted a separate calibration experiment to train the SSVEP decoder, following the same procedure as the offline experiment in the above brain-controlled drone flight experiment. In the HLU experiment, we asked the subjects to complete the cue-following task, which contained ten online blocks and lasted $\sim 6 \text{ h}$. For each block, the online experiment paradigm remained the same as the ErrP calibration experiment. However, the visual feedback of each eight-channel SSVEP epoch was determined by the decoding result of the online memristor chip. We synchronized the onset of visual feedback with brain signals to extract the ErrP epoch. Using the ErrP epoch, one could evaluate the subject's response to the visual feedback of the SSVEP epoch, that is, whether it was correct. We considered the SSVEP epoch to be correctly decoded when the ErrP detector did not identify an ErrP response. Subsequently, we stored the signals of that epoch and used them to update the memristor decoder.

Estimation of brain control state. During co-evolution experiments, we monitored the SSVEP component in the participants' brain signals, considering it to indicate their brain control state. We collected online brain signals to form a test sample set at each update. We used the FBCCA method on this set to predict the sample labels. The accuracy of the FBCCA represented the current brain control state. High accuracy suggested a stable and effective brain control state, whereas low accuracy indicated an ineffective control state. We could observe changes in the brain control state by monitoring the classification accuracy over time. This provided insights into the evolution of brain control ability and its effect on the performance of the co-evolution BCI system.

Estimation of the memristor decoder's effectiveness. We generated a set of test signals to track the memristor decoder's effectiveness throughout the co-evolution experiment. Specifically, the online data for each update are denoted as $x_{n,f}^s$, where n represents the update index, f is the index of target and s is the subject index. We defined the individual template for each subject as

$$X_f^s = \sum_{n=1}^K x_n^s, \quad (11)$$

where K represents the total number of updates. The sin-cos references signals Y_f can be represented as follows:

$$Y_f = \begin{bmatrix} \sin(2\pi f t) \\ \cos(2\pi f t) \\ \vdots \\ \sin(2\pi N_h f t) \\ \cos(2\pi N_h f t) \end{bmatrix}, \quad (12)$$

where f is the stimulation frequency and N_h is the number of harmonics. By applying the source aliasing matrix estimation method⁵⁸, the estimate aliasing matrix $\hat{\Phi}$ can be obtained by minimizing the mean square error between $\tilde{X}_f^s$ and Y_f . The test signal $\tilde{X}_f^s$ can be constructed as follows:

$$\tilde{X}_f^s = \hat{\Phi} Y_f^s + \eta W_{\text{noise}}^m \quad (13)$$

where W_{noise}^m is random Gaussian noise, $m \in 1, 2, \dots, N_a$ and N_a is the number of constructed signals. The parameter η controls the intensity of noise. Subsequently, these test signals were decoded using the memristor decoder obtained from each update, and the decoding accuracy served as an indicator of the decoder's performance at each update. We repeated the above process ten times to obtain statistically meaningful results, resulting in the average accuracy that represents the decoder's effectiveness. In this study, K is 10, N_h was set to 5, N_a was set to 5 and η was set to 1.8.

Human subjects' information. Five subjects (four females), aged between 22 years and 26 years (average, 24.2 years; standard deviation, 1.48 years), participated in the brain-controlled drone flight experiment. Ten subjects (five females), aged between 22 years and 25 years (average, 23.58 years; standard deviation, 1.13 years), participated in the long-term brain–memristor decoder co-evolution experiment. All subjects were solicited to self-report their biological sex. Complying with the Declaration of Helsinki, we fully informed all subjects of all procedures, so they understood all possible consequences of the study, and signed an informed consent form.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Source data are provided with this paper. Other data that support the findings of this study are available from the corresponding authors upon reasonable request.

Code availability

The core code for memristor-based one-step decoding is publicly available via GitHub at <https://github.com/zhengwuliu/Memristor-Co-evolutional-BCI>. Other code that supports the findings of this study is available from the corresponding authors upon reasonable request.

References

- Willett, F. R., Avansino, D. T., Hochberg, L. R., Henderson, J. M. & Shenoy, K. V. High-performance brain-to-text communication via handwriting. *Nature* **593**, 249–254 (2021).
- Anumanchipalli, G. K., Chartier, J. & Chang, E. F. Speech synthesis from neural decoding of spoken sentences. *Nature* **568**, 493–498 (2019).
- Metzger, S. L. et al. A high-performance neuroprosthesis for speech decoding and avatar control. *Nature* **620**, 1037–1046 (2023).
- Willett, F. R. et al. A high-performance speech neuroprosthesis. *Nature* **620**, 1031–1036 (2023).
- Flesher, S. N. et al. A brain–computer interface that evokes tactile sensations improves robotic arm control. *Science* **372**, 831–836 (2021).
- Lorach, H. et al. Walking naturally after spinal cord injury using a brain–spine interface. *Nature* **618**, 126–133 (2023).
- Silversmith, D. B. et al. Plug-and-play control of a brain–computer interface through neural map stabilization. *Nat. Biotechnol.* **39**, 326–335 (2021).
- Chiang, K.-J., Wei, C.-S., Nakanishi, M. & Jung, T.-P. Boosting template-based SSVEP decoding by cross-domain transfer learning. *J. Neural Eng.* **18**, 016002 (2021).
- Pascual-Leone, A., Amedi, A., Fregni, F. & Merabet, L. B. The plastic human brain cortex. *Annu. Rev. Neurosci.* **28**, 377–401 (2005).
- Luppi, A. I. et al. A synergistic core for human brain evolution and cognition. *Nat. Neurosci.* **25**, 771–782 (2022).
- Abbott, L. F. & Nelson, S. B. Synaptic plasticity: taming the beast. *Nat. Neurosci.* **3**, 1178–1183 (2000).
- Sur, M. & Rubenstein, J. L. R. Patterning and plasticity of the cerebral cortex. *Science* **310**, 805–810 (2005).
- Holz, E. M., Botrel, L., Kaufmann, T. & Kübler, A. Long-term independent brain–computer interface home use improves quality of life of a patient in the locked-in state: a case study. *Arch. Phys. Med. Rehabil.* **96**, S16–S26 (2015).
- Sussillo, D., Stavisky, S. D., Kao, J. C., Ryu, S. I. & Shenoy, K. V. Making brain–machine interfaces robust to future neural variability. *Nat. Commun.* **7**, 13749 (2016).
- Wen, S. et al. Rapid adaptation of brain–computer interfaces to new neuronal ensembles or participants via generative modelling. *Nat. Biomed. Eng.* **7**, 546–558 (2023).
- Renart, A. & Machens, C. K. Variability in neural activity and behavior. *Curr. Opin. Neurobiol.* **25**, 211–220 (2014).
- van Praag, H., Kempermann, G. & Gage, F. H. Neural consequences of environmental enrichment. *Nat. Rev. Neurosci.* **1**, 191–198 (2000).
- Wolpaw, J. R. Brain–computer interfaces as new brain output pathways. *J. Physiol.* **579**, 613–619 (2007).
- Musk, E. An integrated brain–machine interface platform with thousands of channels. *J. Med. Internet Res.* **21**, e16194 (2019).
- Sahasrabudde, K. et al. The Argo: a high channel count recording system for neural recording in vivo. *J. Neural Eng.* **18**, 015002 (2021).
- Steinmetz, N. A. et al. Neuropixels 2.0: a miniaturized high-density probe for stable, long-term brain recordings. *Science* **372**, eabf4588 (2021).
- Grover, P. & Venkatesh, P. An information-theoretic view of EEG sensing. *Proc. IEEE* **105**, 367–384 (2016).
- Han, J. et al. A high-speed hybrid brain–computer interface with more than 200 targets. *J. Neural Eng.* **20**, 016025 (2023).
- Chen, Y. et al. Implementing a calibration-free SSVEP-based BCI system with 160 targets. *J. Neural Eng.* **18**, 046094 (2021).
- Horowitz, M. 1.1 Computing's energy problem (and what we can do about it). In *2014 IEEE International Solid-State Circuits Conference Digest of Technical Papers (ISSCC)* 10–14 (IEEE, 2014).
- Dethier, J., Nuyujukian, P., Ryu, S. I., Shenoy, K. V. & Boahen, K. Design and validation of a real-time spiking-neural-network decoder for brain–machine interfaces. *J. Neural Eng.* **10**, 036008 (2013).
- Boi, F. et al. A bidirectional brain–machine interface featuring a neuromorphic hardware decoder. *Front. Neurosci.* **10**, 563 (2016).
- Shaikh, S., So, R., Sibindi, T., Libedinsky, C. & Basu, A. Real-time closed loop neural decoding on a neuromorphic chip. In *2019 9th International IEEE/EMBS Conference on Neural Engineering (NER)* 670–673 (IEEE, 2019).
- Sharifshazileh, M., Burelo, K., Sarnthein, J. & Indiveri, G. An electronic neuromorphic system for real-time detection of high frequency oscillations (HFO) in intracranial EEG. *Nat. Commun.* **12**, 3095 (2021).
- Ambrogio, S. et al. Equivalent-accuracy accelerated neural-network training using analogue memory. *Nature* **558**, 60–67 (2018).
- Wan, W. et al. A compute-in-memory chip based on resistive random-access memory. *Nature* **608**, 504–512 (2022).
- Xia, Q. & Yang, J. J. Memristive crossbar arrays for brain-inspired computing. *Nat. Mater.* **18**, 309–323 (2019).
- Guo, Z. et al. An emerging NVM CIM accelerator with shared-path transpose read and bit-interleaving weight storage for efficient on-chip training in edge devices. In *IEEE Transactions on Circuits and Systems II: Express Briefs* 2645–2649 (IEEE, 2023).

34. Gao, B. et al. Memristor-based analogue computing for brain-inspired sound localization with in situ training. *Nat. Commun.* **13**, 2026 (2022).
35. Zhang, W. et al. Edge learning using a fully integrated neuro-inspired memristor chip. *Science* **381**, 1205–1211 (2023).
36. Liu, Z. et al. Neural signal analysis with memristor arrays towards high-efficiency brain–machine interfaces. *Nat. Commun.* **11**, 4234 (2020).
37. Liu, Z. et al. Multichannel parallel processing of neural signals in memristor arrays. *Sci. Adv.* **6**, eabc4797 (2020).
38. Yuan, R. et al. A neuromorphic physiological signal processing system based on VO₂ memristor for next-generation human–machine interface. *Nat. Commun.* **14**, 3695 (2023).
39. Lee, D. et al. Oxide based nanoscale analog synapse device for neural signal recognition system. In *2015 IEEE International Electron Devices Meeting (IEDM)* 4.7.1–4.7.4 (IEEE, 2015).
40. Park, S. et al. Electronic system with memristive synapses for pattern recognition. *Sci. Rep.* **5**, 10123 (2015).
41. Zhu, X., Wang, Q. & Lu, W. D. Memristor networks for real-time neural activity analysis. *Nat. Commun.* **11**, 2439 (2020).
42. Gupta, I. et al. Real-time encoding and compression of neuronal spikes by metal-oxide memristors. *Nat. Commun.* **7**, 12805 (2016).
43. Chen, X. et al. High-speed spelling with a noninvasive brain–computer interface. *Proc. Natl Acad. Sci. USA* **112**, E6058–E6067 (2015).
44. Mahmood, M. et al. Fully portable and wireless universal brain–machine interfaces enabled by flexible scalp electronics and deep learning algorithm. *Nat. Mach. Intell.* **1**, 412–422 (2019).
45. Chiang, K.-J. et al. A closed-loop adaptive brain–computer interface framework: improving the classifier with the use of error-related potentials. In *2021 10th International IEEE/EMBS Conference on Neural Engineering (NER)* 487–490 (IEEE, 2021).
46. Chavarriaga, R., Sobolewski, A. & Millán, J. D. R. Errare machinale est: the use of error-related potentials in brain–machine interfaces. *Front. Neurosci.* **8**, 208 (2014).
47. Chavarriaga, R. & Millán, J. D. R. Learning from EEG error-related potentials in noninvasive brain–computer interfaces. *IEEE Trans. Neural Syst. Rehabil. Eng.* **18**, 381–388 (2010).
48. Nakanishi, M. et al. Enhancing detection of SSVEPs for a high-speed brain speller using task-related component analysis. *IEEE Trans. Biomed. Eng.* **65**, 104–112 (2018).
49. Gulati, T., Guo, L., Ramanathan, D. S., Bodepudi, A. & Ganguly, K. Neural reactivations during sleep determine network credit assignment. *Nat. Neurosci.* **20**, 1277–1284 (2017).
50. Müller, J. S. et al. A mathematical model for the two-learners problem. *J. Neural Eng.* **14**, 036005 (2017).
51. Orsborn, A. L. et al. Closed-loop decoder adaptation shapes neural plasticity for skillful neuroprosthetic control. *Neuron* **82**, 1380–1393 (2014).
52. Madduri, M. M., Burden, S. A. & Orsborn, A. L. Biosignal-based co-adaptive user–machine interfaces for motor control. *Curr. Opin. Biomed. Eng.* **27**, 100462 (2023).
53. Shanechi, M. M. et al. Rapid control and feedback rates enhance neuroprosthetic control. *Nat. Commun.* **8**, 13825 (2017).
54. Ahmadipour, P., Yang, Y., Chang, E. F. & Shanechi, M. M. Adaptive tracking of human ECoG network dynamics. *J. Neural Eng.* **18**, 016011 (2021).
55. Meidahl, A. C. et al. Adaptive deep brain stimulation for movement disorders: the long road to clinical therapy. *Mov. Disord.* **32**, 810–819 (2017).
56. Shin, U. et al. NeuralTree: a 256-channel 0.227-μJ/class versatile neural activity classification and closed-loop neuromodulation SoC. *IEEE J. Solid-State Circuits* **57**, 3243–3257 (2022).
57. Xiao, X. et al. Discriminative canonical pattern matching for single-trial classification of ERP components. *IEEE Trans. Biomed. Eng.* **67**, 2266–2275 (2019).
58. Luo, R. et al. Data augmentation of SSVEPs using source aliasing matrix estimation for brain–computer interfaces. *IEEE Trans. Biomed. Eng.* **70**, 1775–1785 (2023).

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Author contributions

J.T., M.X., D.M. and H.W. supervised the project. Z.L., J.M., J.T., M.X., D.M. and H.W. conceived the idea. Z.L., J.M., J.T. and M.X. designed the experiments and wrote the paper. Q.L., J.T., B.G., H.Q. and H.W. contributed to the design and fabrication of the memristor chip. Z.L., S.D. and Q.Q. designed the chip test system. Z.L., Y.X. and Y.L. tested and characterized the memristor devices. Z.L., J.M., J.T. and M.X. designed the interactive co-evolution scheme. Z.L. and J.T. designed the memristor-related algorithms and implemented them on the co-evolution BCI system. J.M., M.X. and W.C. designed the co-evolution BCI software at the PC end and implemented the electroencephalogram decoding algorithm. J.M., M.X., K.W. and W.C. designed the BCI paradigm. J.M. collected the brain signals for the simulated online experiments. Z.L. developed and implemented the one-step decoding models on memristor chips. Z.L. and J.M. performed the memristor-chip-based brain-controlled drone flight experiments. Z.L., J.M. and W.C. performed the brain–memristor decoder co-evolution experiments. P.Y., H.Z., N.W., B.H. and T.-P.J. discussed the results and benchmark. All authors commented on the paper.

Competing interests

The authors declare no competing interests.

Additional information

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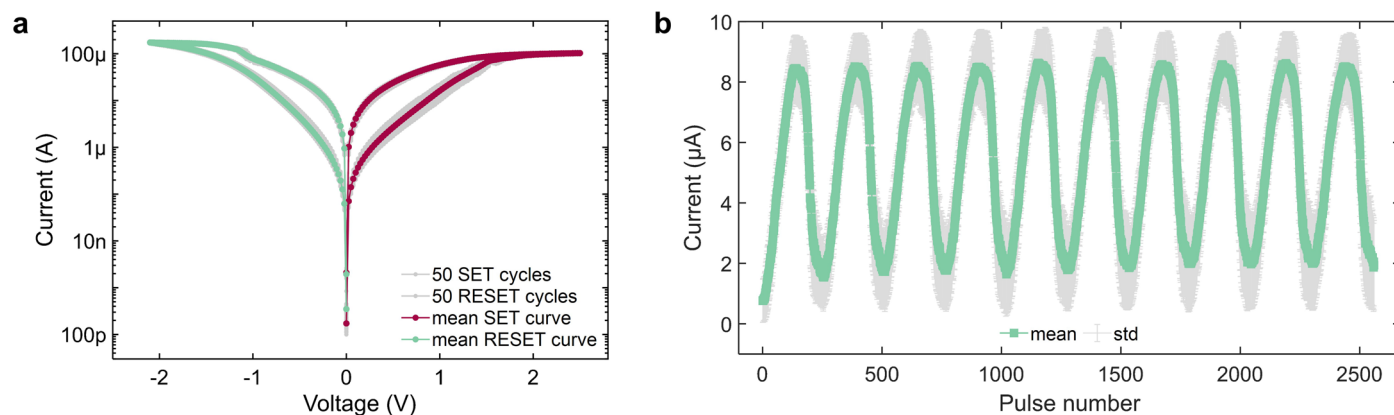
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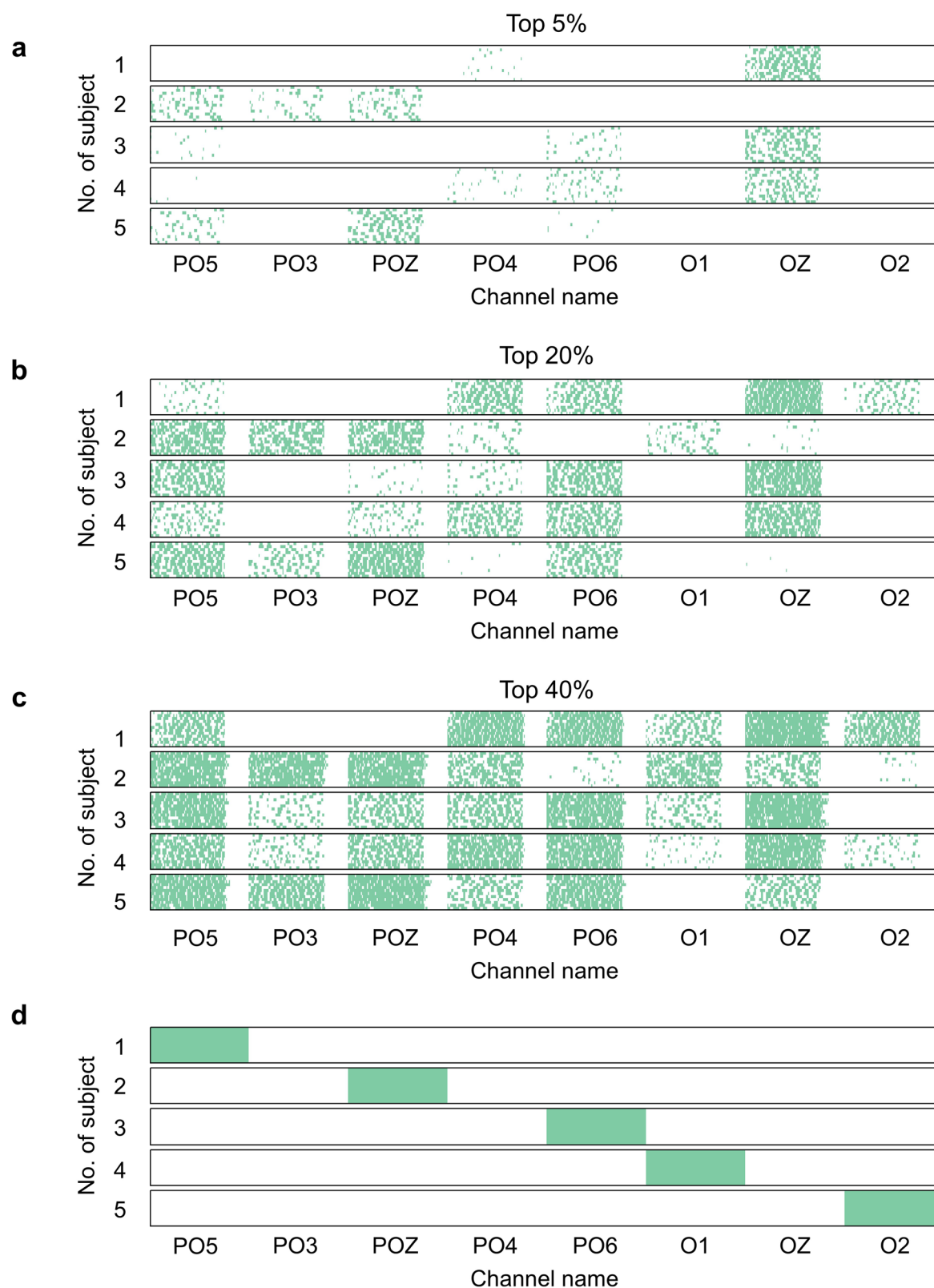
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Extended Data Fig. 1 | Analog resistive switching characteristics of memristors. **a**, Direct-current (DC) I-V curves of a typical memristor device. The red and green lines plot the mean values of 50 consecutive switching cycles, which are indicated by grey lines. **b**, Analog resistive switching characteristics

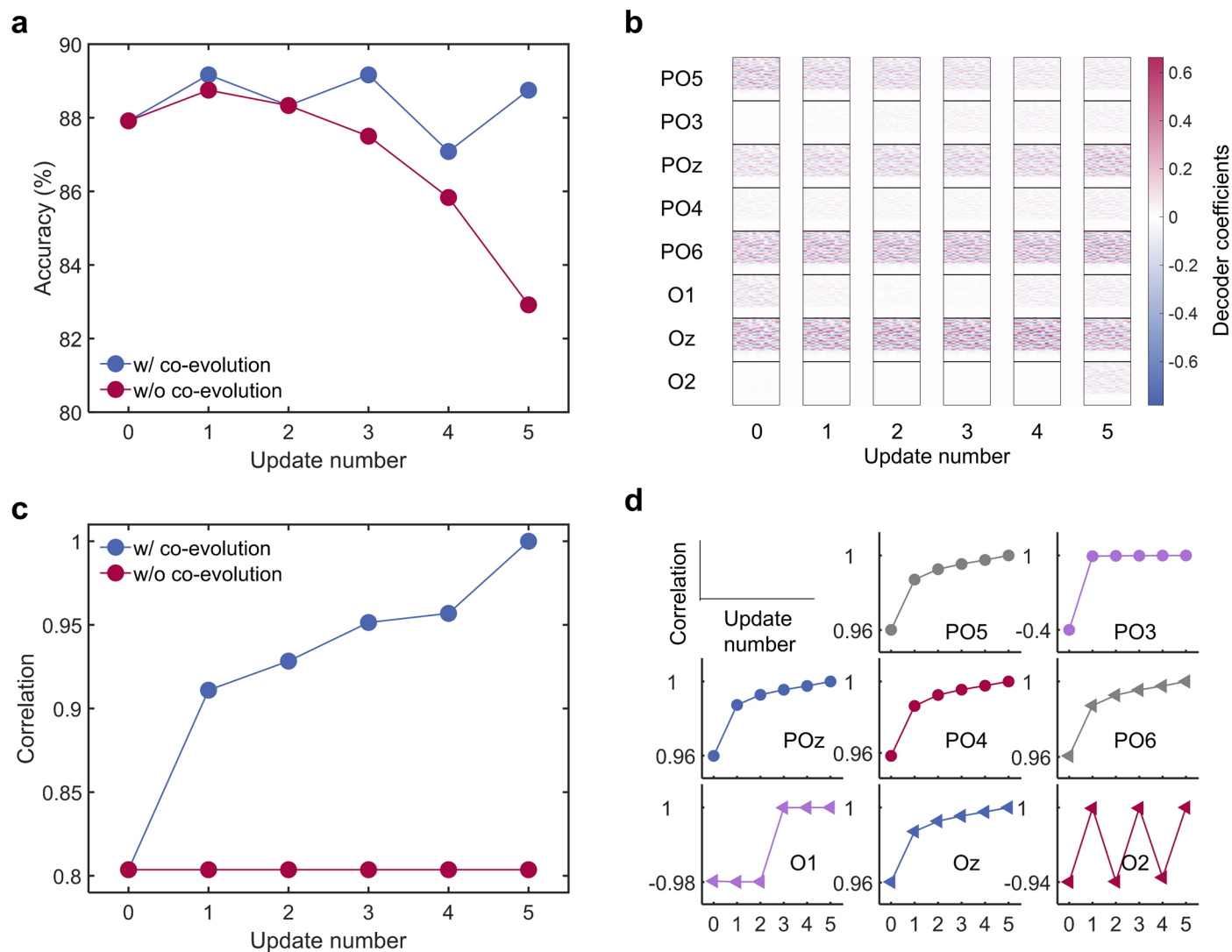
for 128 representative memristor devices under 10 consecutive pulsed SET and RESET cycles. The read voltage is fixed at 0.2 V. Data are presented as mean values $\pm$ s.d.



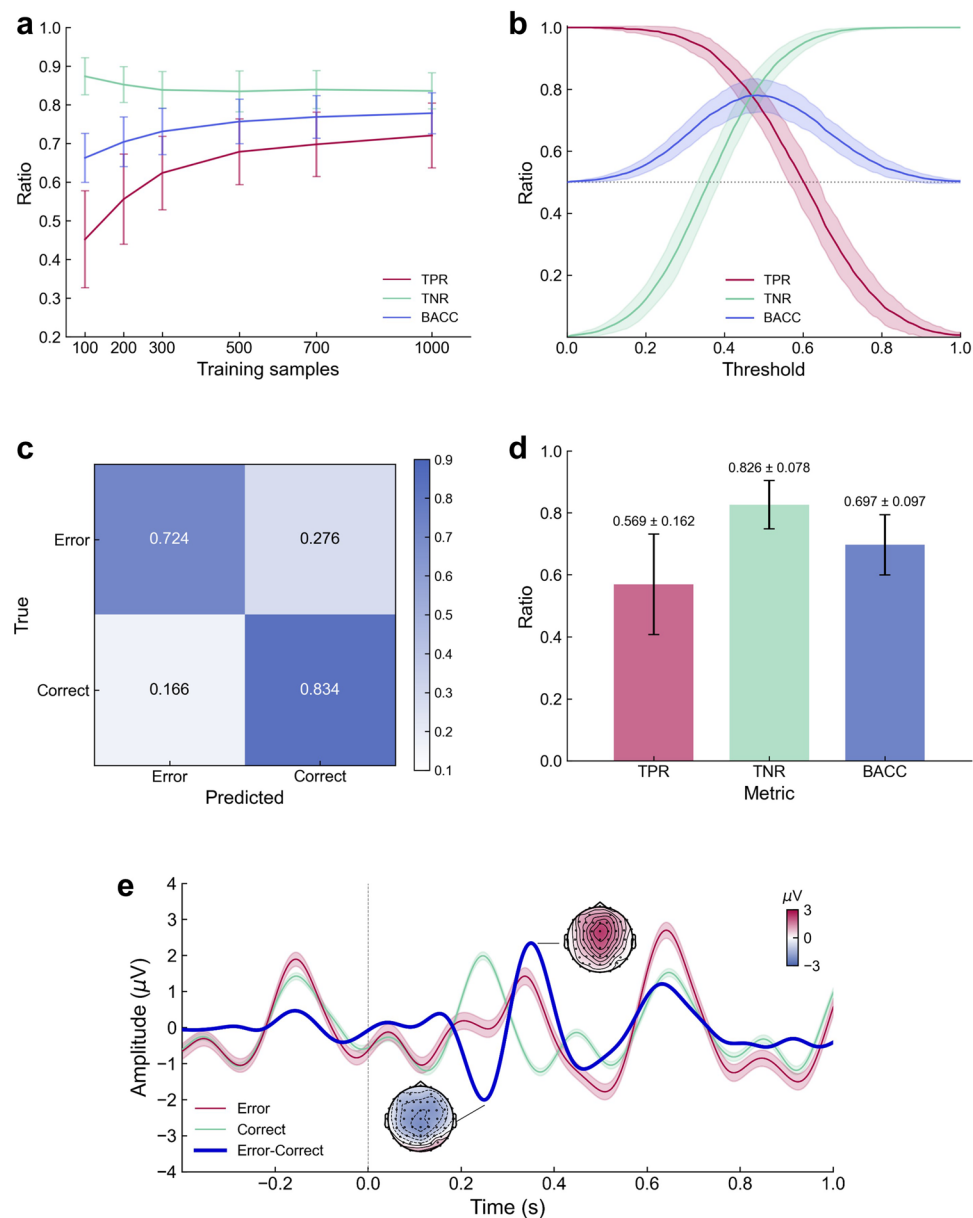
Extended Data Fig. 2 | Binary masks for parameter pruning experiments.

a, Mask for parameters whose magnitudes are at the top 5%, **b**, Mask for parameters whose magnitudes are at the top 20%, **c**, Mask for parameters whose magnitudes

are at the top 40%, **d**, Mask examples for single-channel decoding experiments. Here the activated channels for subjects A1-A5 are channels PO5, POz, PO6, O1 and O2, respectively.

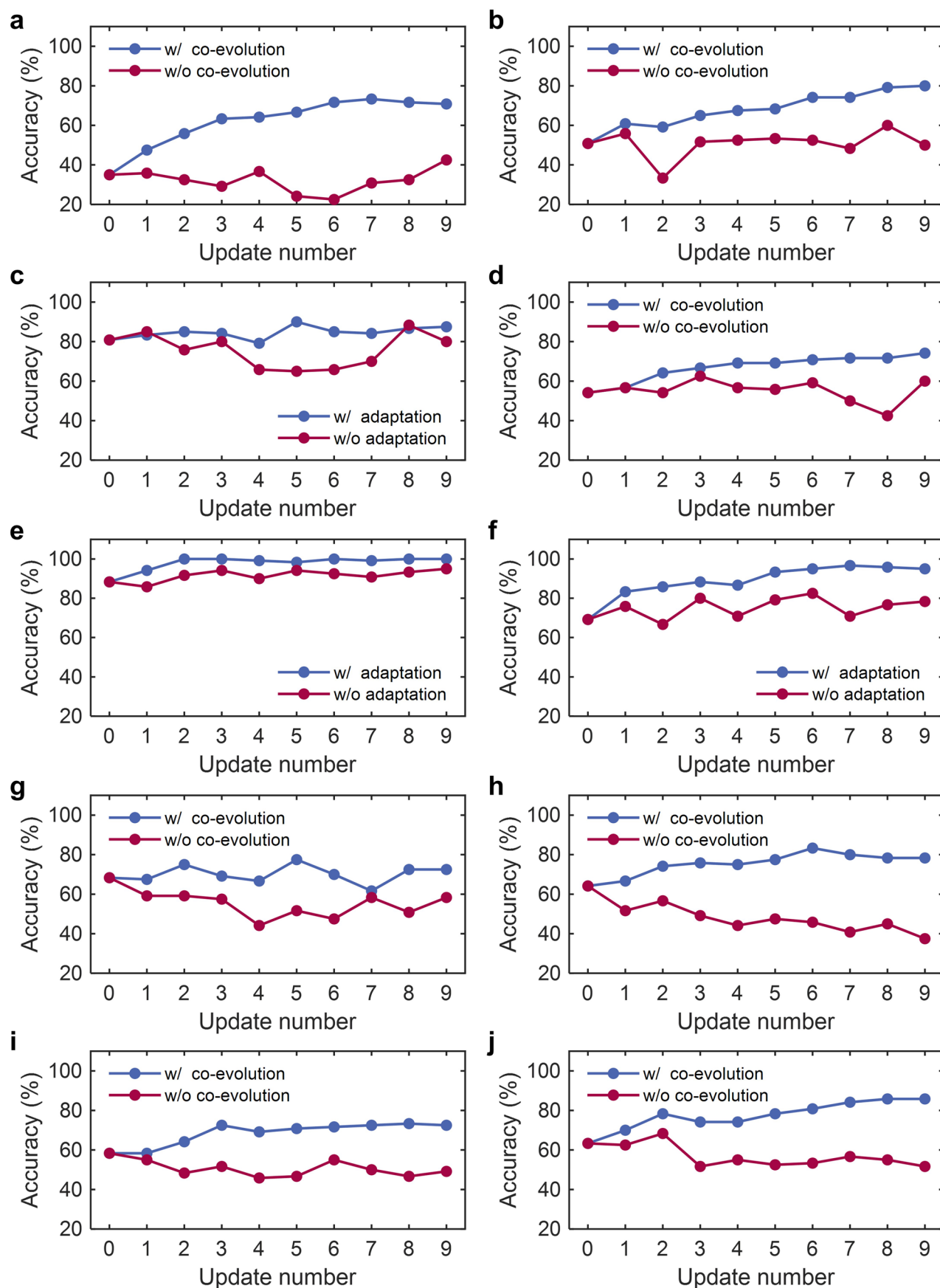


Extended Data Fig. 3 | Results of SU experiments for brain-memristor decoder co-evolution. a, Decoding accuracies over 5 update cycles for the BCIs with (w/) and without (w/o) co-evolution. **b**, Adaptive memristor decoder map after each update cycle. **c**, Correlation coefficients between each decoder map and the final one. **d**, Correlation coefficients between the decoder maps for each channel.



Extended Data Fig. 4 | Overall performance of the ErrP detector. a, Average TPR, TNR and BACC as a function of the number of samples ($n = 100, 200, 300, 500, 700$, and 1000) in the training set, and the testing set contains 360 samples. The samples in the training set and testing set are randomly selected from the data collected in the ErrP calibration experiment without overlap. **b**, Average TPR, TNR and BACC as a function of the classification threshold. If the feature value of the DCPM-based classifier exceeds a predefined threshold, it labels the classification of “target”; otherwise, it labels the classification of “non-target”.

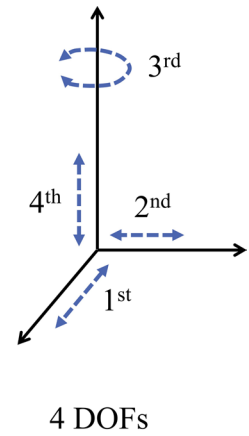
These metrics are calculated from the leave-one-block-out cross-validation on the training data. **c**, Confusion matrix between the “target” class and the “non-target” class when the threshold is set as 0.5 in **b**. **d**, Grand averaged metrics on 10 blocks online data from all 10 subjects ($n = 100$). The error bars in **a** and **d**, and the shaded areas in **b** indicate the mean values $\pm$ s.d. **e**, Correct and incorrect evoked signals from channel FCZ (green and red lines, respectively). The shade areas in **e** indicate the 95% confidence intervals.



Extended Data Fig. 5 | Accuracy evolution for BCIs with and without co-evolution for 10 subjects. a–j. Decoding accuracy results for subjects B1–B10, respectively.

Extended Data Table 1 | Command encoding for the 4-DOF brain-controlled drone flight

No.	DOF	Command	Frequency (Hz)	Phase (π rad)
1	N.A.	Take Off	9.6	1.4
2		Land	11.6	1.15
3	1 st	Forward	9.2	1.05
4	1 st	Back	8.0	0.0
5	2 nd	Left	10.4	0.1
6	2 nd	Right	8.8	0.7
7	3 rd	CW	8.4	0.35
8	3 rd	CCW	10.8	0.45
9	4 th	Up	10.0	1.75
10	4 th	Down	11.2	0.8
11	N.A.	Keep	12.0	1.8
12		Hover	12.4	1.45



DOF: degree of freedom; CW: clockwise; CCW: counterclockwise.

Extended Data Table 2 | Decoding results by the memristor chip and CPU for 5 different subjects

No. of subject	Decoding with CPU		Decoding with memristor	
	Accuracy (%)	ITR (bits/min)	Accuracy (%)	ITR (bits/min)
A1	88.75	134.42	88.75	134.42
A2	83.75	119.13	81.25	112.01
A3	85.42	124.06	85.42	124.06
A4	83.33	117.92	82.08	114.34
A5	89.17	135.77	88.33	133.08
Mean	86.08	126.26	85.17	123.58
Std	2.74	8.40	3.46	10.33

Extended Data Table 3 | Benchmark of energy consumption and area for the memristor chip-based decoder

Components	Area (μm^2)	Energy (pJ)
BL driver	13974.3	120.0
WL driver	345.2	16.5
Memristor array	1297.9	153.6
S&H	1.0	0.1
ADC	28200	840.0
Shift adder	5233.2	30.5
Subtractor	8858.3	66.0
Total	57909.9	1226.7

BL, bit line; WL, word line; S&H, sample & hold; ADC, analog-to-digital converter.

Extended Data Table 4 | Pruning experiments with masks on the decoder parameters

a

No. of subject	Retained parameters				
	Top 20%	Top 40%	Top 60%	Top 80%	Top 100%
1	89.58	90.00	90.00	90.42	90.42
2	79.17	81.67	83.75	83.33	83.33
3	78.75	78.33	77.50	77.50	77.08
4	75.00	80.33	80.42	80.83	80.83
5	87.50	86.67	85.83	85.42	85.41
Mean	82.00	83.40	83.50	83.50	83.41

b

No. of subject	Pruned parameters						
	None	Top 1%	Top 2%	Top 5%	Top 10%	Top 15%	Top 20%
1	90.42	90.00	90.00	90.42	74.17	49.58	30.83
2	83.33	82.92	77.50	69.17	57.92	46.25	41.25
3	77.08	72.92	65.83	49.17	33.75	24.17	19.58
4	80.83	77.08	76.67	71.25	44.17	28.33	17.92
5	85.41	87.50	85.42	83.75	83.33	72.08	55.83
Mean	83.41	82.08	79.08	72.75	58.67	44.08	33.08

a, Results for 5 different experiments where masks are applied to retain parameters with the top 20%, 40%, 60%, 80% and 100% of magnitudes, and prune the others. **b**, Results for 7 different masks that are applied to prune parameters with the top 1%, 2%, 5%, 10%, 15%, 20% and 40% of magnitudes, and retain the others.

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☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
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☐	☒ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted <i>Give P values as exact values whenever suitable.</i>
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Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

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Data collection	The brain data involved in this article were collected using Synamp2 system (Neuroscan Inc.) and its corresponding acquisition software Scan4.5. More details can be found at "Brain signal recording and decoding" in Method.
Data analysis	In the sections "Framework of a memristor chip-enabled BCI" and "Real-time brain-controlled drone flight with memristor chip-based decoder", the code implemented using Matlab software is used to pre-process brain data and complete the simulation experiment. In the section "Brain-memristor decoder co-evolution for enhanced BCI", the offline analysis including SSVEP decoding algorithm and ErrP decoding algorithm was conducted by custom codes in Python. The data analysis of all online experiments in this article was performed using custom codes in Python and C with memristor-based system. All codes that support the findings of this study are available from the corresponding authors upon reasonable request.

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Source data for Figs. 2e-i, Figs. 3d-i and Figs. 4a-h are provided. Other data that support the findings of this study are available from the corresponding authors upon reasonable request.

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Reporting on sex and gender

A total of 15 subjects (five males and ten females). Participants were solicited to self-report their biology sex. Complying with the Declaration of Helsinki, we fully informed all subjects of all procedures, so they understood all possible consequences of the study, and signed an informed consent form.

Reporting on race, ethnicity, or other socially relevant groupings

Five subjects (four females), aged between 22 and 26 (average: 24.2, standard deviation: 1.48), participated in the brain-controlled drone flight experiment. Ten subjects (five females), aged between 22 and 25 (average: 23.58, standard deviation: 1.13), participated in the long-term brain-memristor decoder co-evolution experiment. They are:

A1, female, Age 24, East-Asian, no relevant diagnosis;
 A2, female, Age 24, East-Asian, no relevant diagnosis;
 A3, male, Age 22, East-Asian, no relevant diagnosis;
 A4, female, Age 25, East-Asian, no relevant diagnosis;
 A5, female, Age 26, East-Asian, no relevant diagnosis;
 B1, male, Age 25, East-Asian, no relevant diagnosis;
 B2, female, Age 23, East-Asian, no relevant diagnosis;
 B3, female, Age 22, East-Asian, no relevant diagnosis;
 B4, male, Age 23, East-Asian, no relevant diagnosis;
 B5, female, Age 24, East-Asian, no relevant diagnosis;
 B6, male, Age 24, East-Asian, no relevant diagnosis;
 B7, female, Age 25, East-Asian, no relevant diagnosis;
 B8, female, Age 24, East-Asian, no relevant diagnosis;
 B9, male, Age 22, East-Asian, no relevant diagnosis;
 B10, male, Age 24, East-Asian, no relevant diagnosis;

Population characteristics

All subjects were healthy students or employees of Tianjin University, and they had no known diseases that might affect the experiments.

Recruitment

We published a recruitment advertisement for subjects to students at Tianjin University and recruited subjects in the order of registration. All subjects voluntarily participated in the experiments involved in this study.

Ethics oversight

The experimental protocol was approved by the Institutional Review Board at Tianjin University.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

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Life sciences study design

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Sample size

No statistical methods were used to predetermine the sample size because this study is a proof-of-concept investigation. The chosen sample size aligns with commonly accepted ranges in non-invasive BCI-related experiments, which typically include 10 to 20 participants.

Data exclusions

No data were excluded.

Replication

In the two main experiments, 5 and 10 subjects respectively completed the same experimental task, representing 5 and 10 independent

Replication	replications with different participants. The individual results, as shown in the Extended Data, demonstrate strong inter-subject consistency, supporting the reproducibility of our experimental findings across different individuals.
Randomization	Randomization was not applicable to this study because all subjects participated in the same experimental tasks. The aim of this study was to investigate the functionality of memristors in decoding EEG signals, assess their performance characteristics, and analyze group-level trends across all participants as the brain and memristors co-evolved.
Blinding	Blinding was not implemented in this study because there was no grouping, and all subjects performed the same experimental task. The comparison results were not influenced by experimenter or data analyst preferences, as they were solely determined by the application of different methods to the same set of data.

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Methods

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Authentication	Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosaicism, off-target gene editing) were examined.